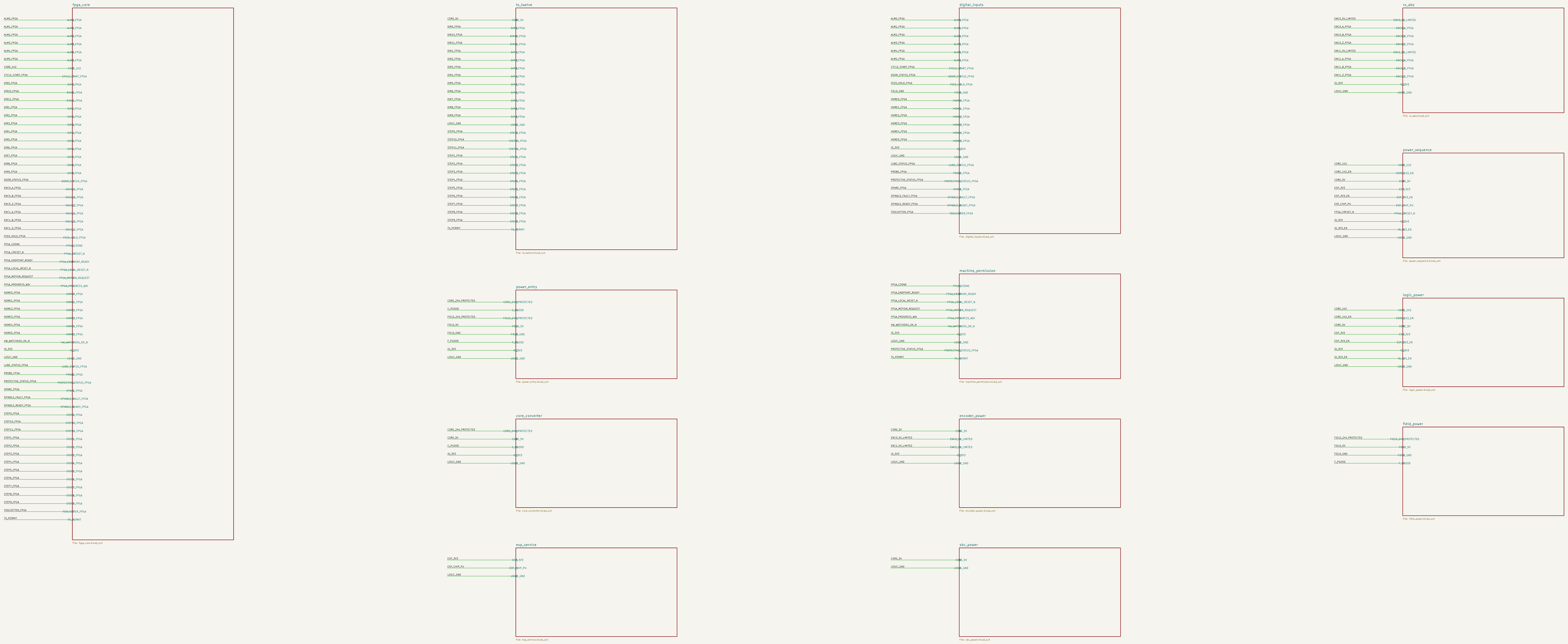
CONNECTED CONTROLLER CIRCUIT STUDY — TWELVE STEP/DIR CHANNELS / SIX—CHANNEL RUNTIME

Real subcircuits connected through named sheet ports. Independent LOGIC / FIELD returns.
Compute carrier, remaining field outputs/drive interfaces and analog circuits remain unfinished. Expanded power and firmware are not qualified.
Boundary connectors are retained review interfaces; connector selection and full PCB layout remain open.


fpga_core
ALM0_FPGA
ALM1_FPGA
ALM2_FPGA
ALM3_FPGA
ALM4_FPGA
ALM5_FPGA
CORE_1V2
CYCLE_START_FPGA
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
DOOR_STATUS_FPGA
ENC0_A_FPGA
ENC0_B_FPGA
ENC0_Z_FPGA
ENC1_A_FPGA
ENC1_B_FPGA
ENC1_Z_FPGA
FEED_HOLD_FPGA
FPGA_CDONE
FPGA_CRESET_N
FPGA_ENDPOINT_READY
FPGA_LOCAL_RESET_N
FPGA_MOTION_REQUEST
FPGA_PROGRESS_WDI
HOME0_FPGA
HOME1_FPGA
HOME2_FPGA
HOME3_FPGA
HOME4_FPGA
HOME5_FPGA
HW_WATCHDOG_OK_N
IO_3V3
LOGIC_GND
LUBE_STATUS_FPGA
PROBE_FPGA
PROTECTIVE_STATUS_FPGA
SPARE_FPGA
SPINDLE_FAULT_FPGA
SPINDLE_READY_FPGA
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TOOLSETTER_FPGA
TX_PERMIT
File: fpga_core.kicad_sch
tx_twelve
CORE_5V
DIR0_FPGA
DIR10_FPGA
DIR11_FPGA
DIR1_FPGA
DIR2_FPGA
DIR3_FPGA
DIR4_FPGA
DIR5_FPGA
DIR6_FPGA
DIR7_FPGA
DIR8_FPGA
DIR9_FPGA
LOGIC_GND
STEP0_FPGA
STEP10_FPGA
STEP11_FPGA
STEP1_FPGA
STEP2_FPGA
STEP3_FPGA
STEP4_FPGA
STEP5_FPGA
STEP6_FPGA
STEP7_FPGA
STEP8_FPGA
STEP9_FPGA
TX_PERMIT
File: tx_twelve.kicad_sch
power_entry
CORE_24V_PROTECTED
C_PGOOD
FIELD_24V_PROTECTED
FIELD_5V
FIELD_GND
F_PGOOD
IO_3V3
LOGIC_GND
File: power_entry.kicad_sch
core_converter
CORE_24V_PROTECTED
CORE_5V
C_PGOOD
IO_3V3
LOGIC_GND
File: core_converter.kicad_sch
esp_service
ESP_3V3
ESP_CHIP_PU
LOGIC_GND
File: esp_service.kicad_sch
digital_inputs
ALM0_FPGA
ALM1_FPGA
ALM2_FPGA
ALM3_FPGA
ALM4_FPGA
ALM5_FPGA
CYCLE_START_FPGA
DOOR_STATUS_FPGA
FEED_HOLD_FPGA
FIELD_GND
HOME0_FPGA
HOME1_FPGA
HOME2_FPGA
HOME3_FPGA
HOME4_FPGA
HOME5_FPGA
IO_3V3
LOGIC_GND
LUBE_STATUS_FPGA
PROBE_FPGA
PROTECTIVE_STATUS_FPGA
SPARE_FPGA
SPINDLE_FAULT_FPGA
SPINDLE_READY_FPGA
TOOLSETTER_FPGA
File: digital_inputs.kicad_sch
machine_permission
FPGA_CDONE
FPGA_ENDPOINT_READY
FPGA_LOCAL_RESET_N
FPGA_MOTION_REQUEST
FPGA_PROGRESS_WDI
HW_WATCHDOG_OK_N
IO_3V3
LOGIC_GND
PROTECTIVE_STATUS_FPGA
TX_PERMIT
File: machine_permission.kicad_sch
encoder_power
CORE_5V
ENC0_5V_LIMITED
ENC1_5V_LIMITED
IO_3V3
LOGIC_GND
File: encoder_power.kicad_sch
sbc_power
CORE_5V
LOGIC_GND
File: sbc_power.kicad_sch
rx_abz
ENC0_5V_LIMITED
ENC0_A_FPGA
ENC0_B_FPGA
ENC0_Z_FPGA
ENC1_5V_LIMITED
ENC1_A_FPGA
ENC1_B_FPGA
ENC1_Z_FPGA
IO_3V3
LOGIC_GND
File: rx_abz.kicad_sch
power_sequence
CORE_1V2
CORE_1V2_EN
CORE_5V
ESP_3V3
ESP_3V3_EN
ESP_CHIP_PU
FPGA_CRESET_N
IO_3V3
IO_3V3_EN
LOGIC_GND
File: power_sequence.kicad_sch
logic_power
CORE_1V2
CORE_1V2_EN
CORE_5V
ESP_3V3
ESP_3V3_EN
IO_3V3
IO_3V3_EN
LOGIC_GND
File: logic_power.kicad_sch
field_power
FIELD_24V_PROTECTED
FIELD_5V
FIELD_GND
F_PGOOD
File: field_power.kicad_sch

Sheet: /
File: controller_assembly.kicad_sch

Title: Controller connected circuit review

Size: A0	Date: 2026–09–28	**Rev: r0.1**
KiCad E.D.A. 10.0.4		Id: 1/14

DUAL DC POWER ENTRY — PARTIAL ENGINEERING CIRCUIT r0.2

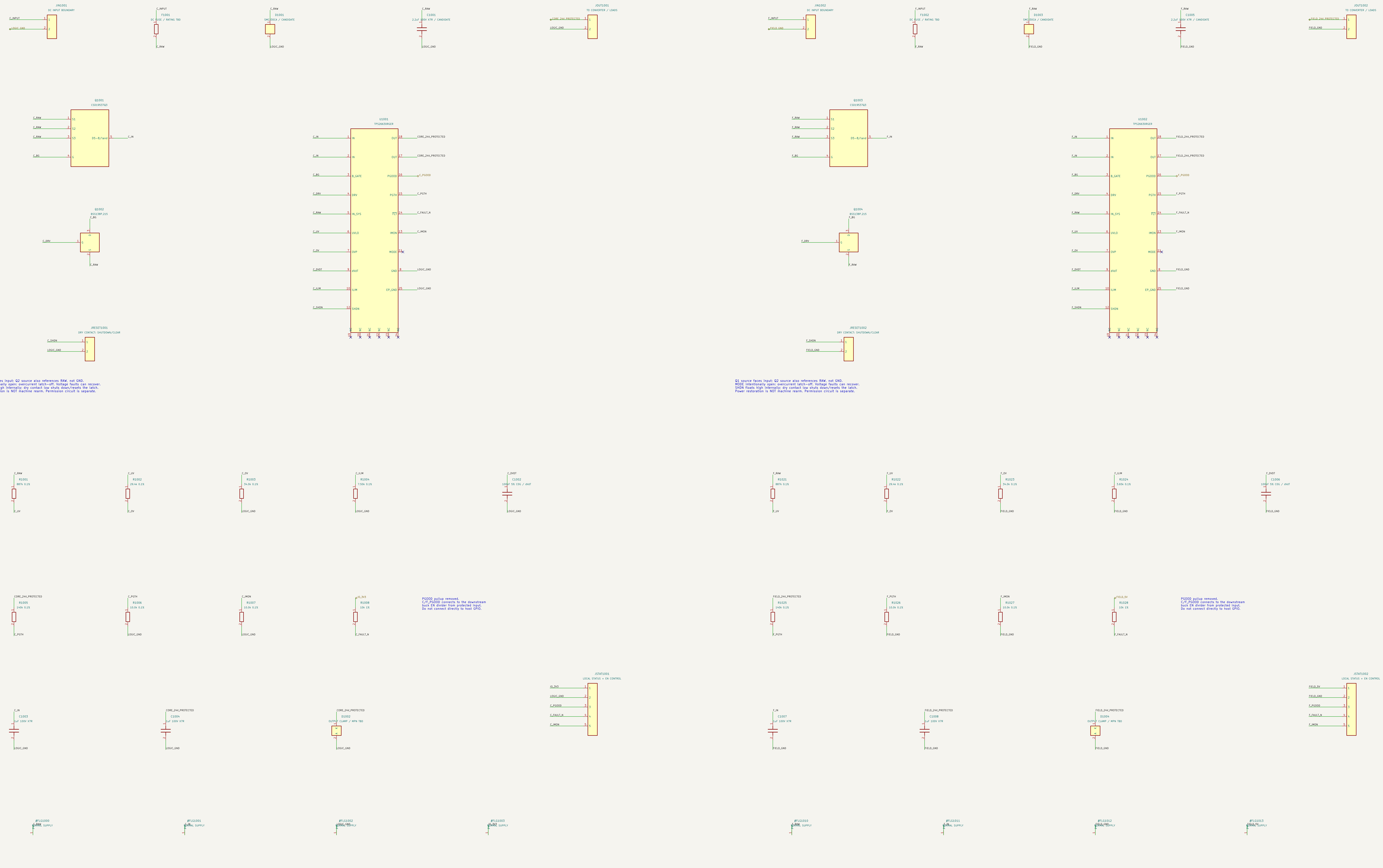
TPS26630 + reverse—blocking FETs; separate LOGIC/FIELD grounds. No PCB or manufacturing release. Exact ratings require the companion review.

CORE / LOGIC DOMAIN

Draft19—30V entry; normal load 1.81A; input path target>=3A.
Display/AI/motor power excluded. Fuse and wiring selection remains open.

FIELD DOMAIN — KEEP ISOLATED

Draft19—30V entry; normal load 4.36A; input path target>=6A.
Display/AI/motor power excluded. Fuse and wiring selection remains open.


JIN1001
DC INPUT BOUNDARY
C_INPUT
LOGIC_GND
F1001
DC FUSE / RATING TBD
C_RAW
D1001
SMCJ30CA / CANDIDATE
C1001
2.2uF 100V X7R / CANDIDATE
JOUT1001
TO CONVERTER / LOADS
CORE_24V_PROTECTED
Q1001
CSD19537Q3
S1
S2
S3
G
D5—8/land
C_BG
C_IN
Q1002
BSS138P,215
C_DRV
D
S
U1001
TPS26630RGER
IN
B_GATE
DRV
IN_SYS
UVLO
OVP
dVdT
ILIM
SHDN
OUT
PGOOD
PGTH
FLT
IMON
MODE
GND
EP_GND
NC
C_UV
C_OV
C_DVDT
C_ILIM
C_SHDN
C_PGOOD
C_PGTH
C_FAULT_N
C_IMON
JRESET1001
DRY CONTACT: SHUTDOWN/CLEAR
R1001
887k 0.1%
R1002
29.4k 0.1%
R1003
34.0k 0.1%
R1004
7.50k 0.1%
C1002
100nF 5% C0G / dVdT
R1005
140k 0.1%
R1006
10.0k 0.1%
R1007
10.0k 0.1%
R1008
10k 1%
IO_3V3
C1003
1uF 100V X7R
C1004
1uF 100V X7R
D1002
OUTPUT CLAMP / MPN TBD
A
K
JSTAT1001
LOCAL STATUS + EN CONTROL
#FLG1000
#FLG1001
#FLG1002
#FLG1003
NORMAL SUPPLY
JIN1002
F_INPUT
FIELD_GND
F1002
F_RAW
D1003
C1005
JOUT1002
FIELD_24V_PROTECTED
Q1003
Q1004
F_BG
F_IN
F_DRV
U1002
F_UV
F_OV
F_DVDT
F_ILIM
F_SHDN
F_PGOOD
F_PGTH
F_FAULT_N
F_IMON
JRESET1002
R1021
887k 0.1%
R1022
29.4k 0.1%
R1023
34.0k 0.1%
R1024
3.65k 0.1%
C1006
100nF 5% C0G / dVdT
R1025
140k 0.1%
R1026
10.0k 0.1%
R1027
10.0k 0.1%
R1028
10k 1%
FIELD_5V
C1007
1uF 100V X7R
C1008
1uF 100V X7R
D1004
JSTAT1002
#FLG1010
#FLG1011
#FLG1012
#FLG1013

Q1 source faces input; Q2 source also references RAW, not GND.
MODE intentionally open: overcurrent latch—off. Voltage faults can recover.
SHDN floats high internally; dry contact low shuts down/resets the latch.
Power restoration is NOT machine rearm. Permission circuit is separate.

PGOOD pullup removed.
C/F_PGOOD connects to the downstream
buck EN divider from protected input.
Do not connect directly to host GPIO.

Fault pullups only use downstream diagnostic supply; PGOOD uses converter EN bias.
Effective RAW bypass >=1uF; IN/OUT >=0.1uF. Candidate MPN/bias curves unresolved.
Total downstream capacitance <=1000uF; field loads require independent inhibit.
TVS/clamp/fuse/thermal qualification open. Power flags declare normal paths only.

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MODE intentionally open: overcurrent latch—off. Voltage faults can recover.
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Total downstream capacitance <=1000uF; field loads require independent inhibit.
TVS/clamp/fuse/thermal qualification open. Power flags declare normal paths only.

Partial circuit; physical fuse/clamp selection and qualification open

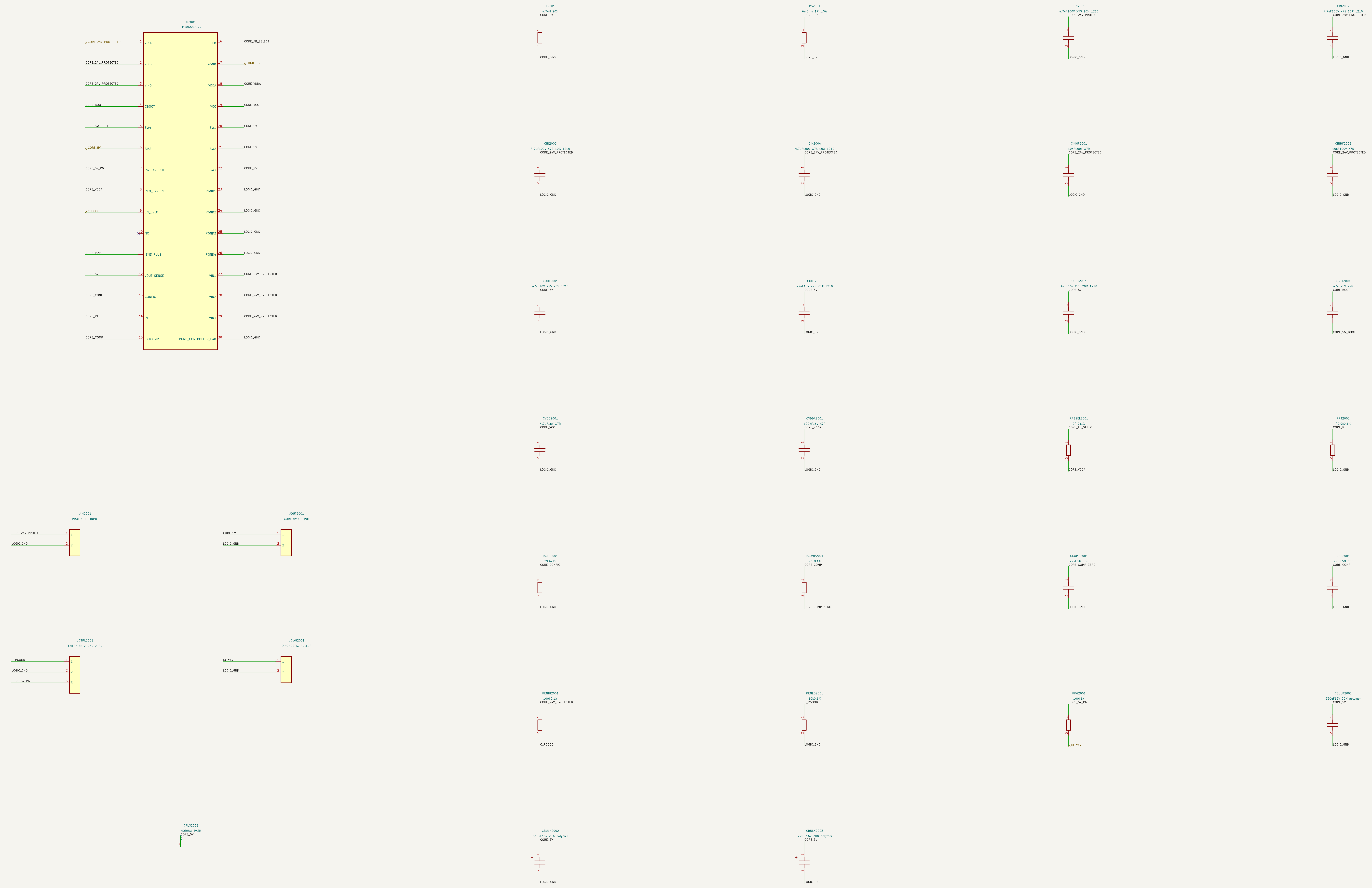
Sheet: /power_entry/
File: power_entry.kicad_sch

Title: Core and field DC power entry

Size: A1
Date: 2026—09—28
Rev: r0.2
KiCad E.D.A. 10.0.4
Id: 2/14

CORE 5 V / 6 A CONVERTER CANDIDATE — ENGINEERING CIRCUIT

LM70660: protected nominal24V input; no display/AI/drive power. C_PGOOD is analog enable, not GPIO.
Upstream R9 must be absent. VIN powers internal startup bias; no running host/ESP is needed to enable this converter.


U2001
LM70660RRXR
CORE_24V_PROTECTED 1 VIN4
CORE_24V_PROTECTED 2 VIN5
CORE_24V_PROTECTED 3 VIN6
CORE_BOOT 4 CBOOT
CORE_SW_BOOT 5 SW4
CORE_5V 6 BIAS
CORE_5V_PG 7 PG_SYNCOUT
CORE_VDDA 8 PFM_SYNCIN
C_PGOOD 9 EN_UVLO
10 NC
CORE_ISNS 11 ISNS_PLUS
CORE_5V 12 VOUT_SENSE
CORE_CONFIG 13 CONFIG
CORE_RT 14 RT
CORE_COMP 15 EXTCOMP
FB 16 CORE_FB_SELECT
AGND 17 LOGIC_GND
VDDA 18 CORE_VDDA
VCC 19 CORE_VCC
SW1 20 CORE_SW
SW2 21 CORE_SW
SW3 22 CORE_SW
PGND1 23 LOGIC_GND
PGND2 24 LOGIC_GND
PGND3 25 LOGIC_GND
PGND4 26 LOGIC_GND
VIN1 27 CORE_24V_PROTECTED
VIN2 28 CORE_24V_PROTECTED
VIN3 29 CORE_24V_PROTECTED
PGND_CONTROLLER_PAD 30 LOGIC_GND
L2001
4.7uH 20%
CORE_SW
CORE_ISNS
RS2001
6mOhm 1% 1.5W
CORE_ISNS
CORE_5V
CIN2001
4.7uF100V X7S 10% 1210
CORE_24V_PROTECTED
LOGIC_GND
CIN2002
4.7uF100V X7S 10% 1210
CORE_24V_PROTECTED
LOGIC_GND
CIN2003
4.7uF100V X7S 10% 1210
CORE_24V_PROTECTED
LOGIC_GND
CIN2004
4.7uF100V X7S 10% 1210
CORE_24V_PROTECTED
LOGIC_GND
CINHF2001
10nF100V X7R
CORE_24V_PROTECTED
LOGIC_GND
CINHF2002
10nF100V X7R
CORE_24V_PROTECTED
LOGIC_GND
COUT2001
47uF10V X7S 20% 1210
CORE_5V
LOGIC_GND
COUT2002
47uF10V X7S 20% 1210
CORE_5V
LOGIC_GND
COUT2003
47uF10V X7S 20% 1210
CORE_5V
LOGIC_GND
CBST2001
47nF25V X7R
CORE_BOOT
CORE_SW_BOOT
CVCC2001
4.7uF16V X7R
CORE_VCC
LOGIC_GND
CVDDA2001
100nF16V X7R
CORE_VDDA
LOGIC_GND
RFBSEL2001
24.9k1%
CORE_FB_SELECT
CORE_VDDA
RRT2001
49.9k0.1%
CORE_RT
LOGIC_GND
JIN2001
PROTECTED INPUT
CORE_24V_PROTECTED 1
LOGIC_GND 2
JOUT2001
CORE 5V OUTPUT
CORE_5V 1
LOGIC_GND 2
RCFG2001
29.4k1%
CORE_CONFIG
LOGIC_GND
RCOMP2001
9.53k1%
CORE_COMP
CORE_COMP_ZERO
CCOMP2001
22nF5% C0G
CORE_COMP_ZERO
LOGIC_GND
CHF2001
330pF5% C0G
CORE_COMP
LOGIC_GND
JCTRL2001
ENTRY EN / GND / PG
C_PGOOD 1
LOGIC_GND 2
CORE_5V_PG 3
JDIAG2001
DIAGNOSTIC PULLUP
IO_3V3 1
LOGIC_GND 2
RENHI2001
100k0.1%
CORE_24V_PROTECTED
C_PGOOD
RENLO2001
10k0.1%
C_PGOOD
LOGIC_GND
RPG2001
100k1%
CORE_5V_PG
IO_3V3
CBULK2001
330uF16V 20% polymer
CORE_5V
LOGIC_GND
#FLG2002
NORMAL PATH
CORE_5V
CBULK2002
330uF16V 20% polymer
CORE_5V
LOGIC_GND
CBULK2003
330uF16V 20% polymer
CORE_5V
LOGIC_GND

Layout: Kelvin ISNS+/VOUT across RS1; separate quiet AGND returns join local power ground as TI specifies.
SW4 bootstrap return is internally common with SW1–3; preserve local bootstrap loop. No external long SW4 strap.
U1 RRX, RS1 long–side0612 and polarized bulk footprints unassigned; no generic substitutions.
Bulk + three MLCCs require delayed SBC branch and revised compensation. Transient margin is conditional.
Compensation, negative clamps and reverse feed remain unqualified. No finished PCB or machine permission.

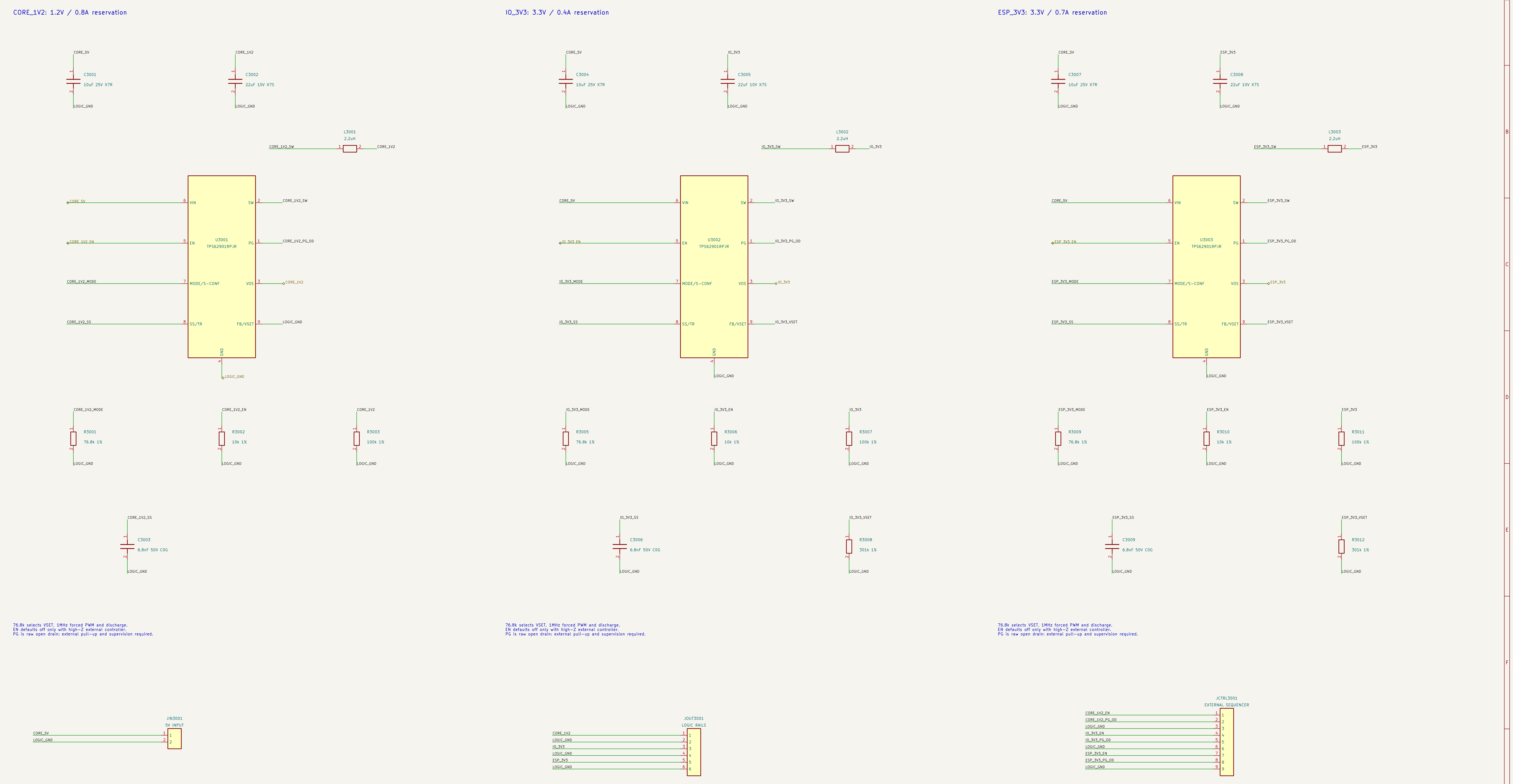
Sheet: /core_converter/
File: core_converter.kicad_sch

Title: Core 5V converter candidate

Size: A1 | Date: 2026–09–28 | **Rev: r0.1**
KiCad E.D.A. 10.0.4 | Id: 3/14

THREE LOW–VOLTAGE REGULATORS – EXTERNAL SEQUENCING REQUIRED

TPS62901 replaces the TPS62160 study here: tighter regulation and adjustable ramp. Frozen ledger unchanged. CORE_5V required 4.85-5.15V.


CORE_1V2: 1.2V / 0.8A reservation
IO_3V3: 3.3V / 0.4A reservation
ESP_3V3: 3.3V / 0.7A reservation
C3001
10uF 25V X7R
C3002
22uF 10V X7S
C3004
10uF 25V X7R
C3005
22uF 10V X7S
C3007
10uF 25V X7R
C3008
22uF 10V X7S
L3001
2.2uH
L3002
2.2uH
L3003
2.2uH
U3001
TPS62901RPJR
U3002
TPS62901RPJR
U3003
TPS62901RPJR
VIN
EN
MODE/S–CONF
SS/TR
SW
PG
VOS
FB/VSET
GND
CORE_5V
CORE_1V2_EN
CORE_1V2_MODE
CORE_1V2_SS
CORE_1V2_SW
CORE_1V2_PG_OD
CORE_1V2
LOGIC_GND
IO_3V3_EN
IO_3V3_MODE
IO_3V3_SS
IO_3V3_SW
IO_3V3_PG_OD
IO_3V3
IO_3V3_VSET
ESP_3V3_EN
ESP_3V3_MODE
ESP_3V3_SS
ESP_3V3_SW
ESP_3V3_PG_OD
ESP_3V3
ESP_3V3_VSET
R3001
76.8k 1%
R3002
10k 1%
R3003
100k 1%
R3005
76.8k 1%
R3006
10k 1%
R3007
100k 1%
R3009
76.8k 1%
R3010
10k 1%
R3011
100k 1%
C3003
6.8nF 50V C0G
C3006
6.8nF 50V C0G
C3009
6.8nF 50V C0G
R3008
301k 1%
R3012
301k 1%
JIN3001
5V INPUT
JOUT3001
LOGIC RAILS
JCTRL3001
EXTERNAL SEQUENCER

76.8k selects VSET, 1MHz forced PWM and discharge.
EN defaults off only with high–Z external controller.
PG is raw open drain: external pull–up and supervision required.

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EN defaults off only with high–Z external controller.
PG is raw open drain: external pull–up and supervision required.

76.8k selects VSET, 1MHz forced PWM and discharge.
EN defaults off only with high–Z external controller.
PG is raw open drain: external pull–up and supervision required.

Discharge is conditional: requires a previous enable and input supply near 2V or above.
No guaranteed power–off discharge, latch–off, reverse–current blocking or machine permission.
ESP rail is separately regulated; independent upstream fault isolation is NOT implemented.

No PG–to–EN cascade. PG thresholds do not prove strict FPGA rail windows; PG is undefined below 1.8V VIN.
6.8nF soft start models 2.231ms nominal; loaded ramp, input fault and restart sequencing remain unmeasured.
VCC/PLL must precede VCCIO; FPGA reset and package supply/filter/decoupling connections are outside this block.

Engineering source; no PCB or physical qualification

Sheet: /logic_power/
File: logic_power.kicad_sch

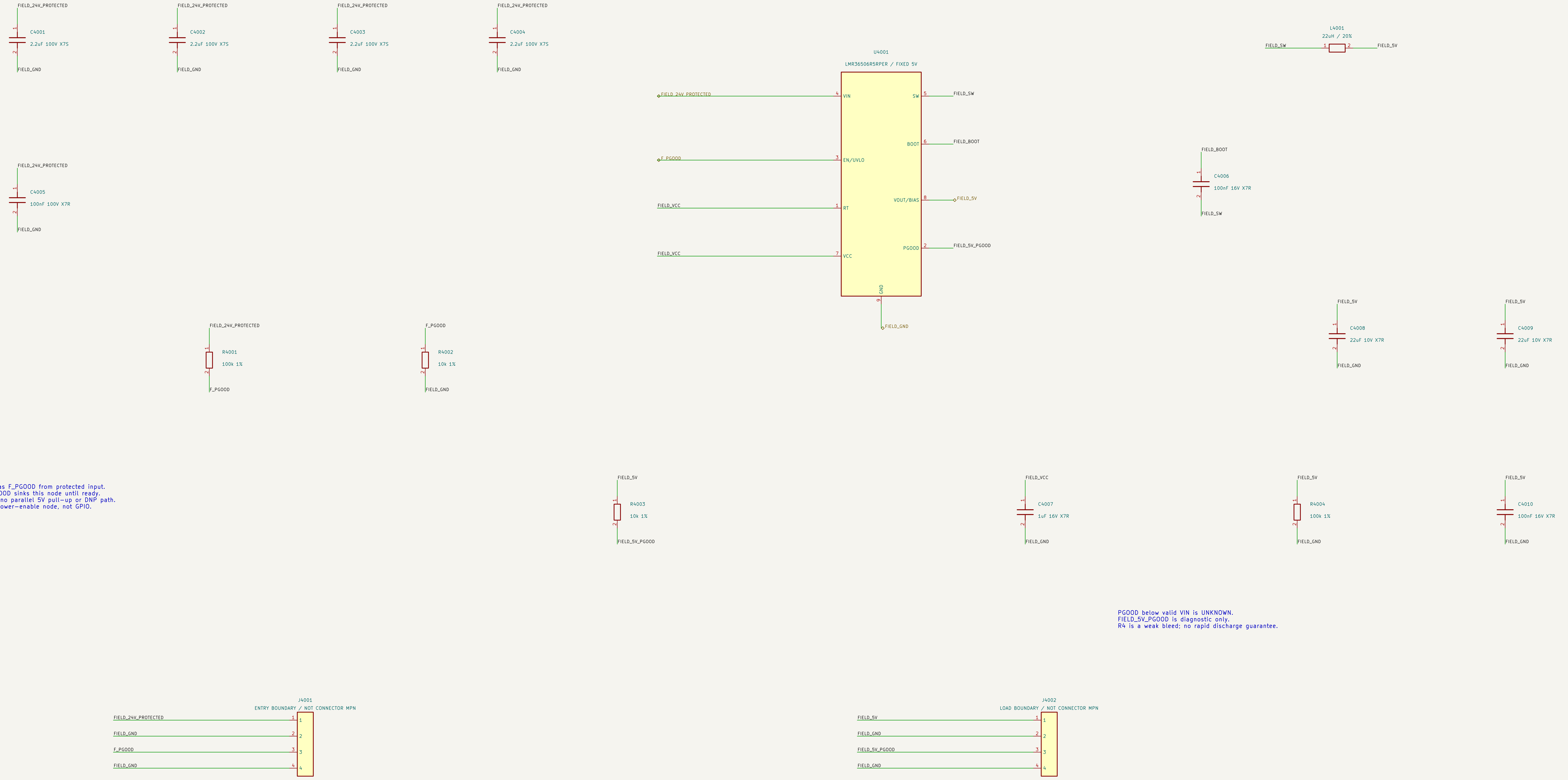
Title: Baseline logic power regulator block

Size: A2
Date: 2026–09–28
Rev: r0.1
KiCad E.D.A. 10.0.4
Id: 4/14

FIELD 24V −> 5V / 100mA − STANDALONE REVIEW CIRCUIT r0.1

All returns are FIELD_GND. This buck is NOT an isolation barrier. No LOGIC_GND connection.
18−30V local−input study; entry chain currently proposes 19−30V. Negative input / retained−output backfeed NOT qualified.

Input effective capacitance >=4.7uF is a qualification gate, not proved by nominal sum. C1−C5 close to VIN/GND.
U1 RT tied to internal VCC selects 1MHz. VCC powers no external load; C6 connects BOOT to SW, never ground.


FIELD_24V_PROTECTED
C4001
2.2uF 100V X7S
FIELD_GND
FIELD_24V_PROTECTED
C4002
2.2uF 100V X7S
FIELD_GND
FIELD_24V_PROTECTED
C4003
2.2uF 100V X7S
FIELD_GND
FIELD_24V_PROTECTED
C4004
2.2uF 100V X7S
FIELD_GND
FIELD_24V_PROTECTED
C4005
100nF 100V X7R
FIELD_GND
U4001
LMR36506R5RPER / FIXED 5V
FIELD_24V_PROTECTED
VIN
F_PGOOD
EN/UVLO
FIELD_VCC
RT
FIELD_VCC
VCC
SW
FIELD_SW
BOOT
FIELD_BOOT
VOUT/BIAS
FIELD_5V
PGOOD
FIELD_5V_PGOOD
GND
FIELD_GND
L4001
22uH / 20%
FIELD_SW
FIELD_5V
FIELD_BOOT
C4006
100nF 16V X7R
FIELD_SW
FIELD_5V
C4008
22uF 10V X7R
FIELD_GND
FIELD_5V
C4009
22uF 10V X7R
FIELD_GND
FIELD_24V_PROTECTED
R4001
100k 1%
F_PGOOD
F_PGOOD
R4002
10k 1%
FIELD_GND
FIELD_5V
R4003
10k 1%
FIELD_5V_PGOOD
FIELD_VCC
C4007
1uF 16V X7R
FIELD_GND
FIELD_5V
R4004
100k 1%
FIELD_GND
FIELD_5V
C4010
100nF 16V X7R
FIELD_GND
J4001
ENTRY BOUNDARY / NOT CONNECTOR MPN
FIELD_24V_PROTECTED
FIELD_GND
F_PGOOD
FIELD_GND
J4002
LOAD BOUNDARY / NOT CONNECTOR MPN
FIELD_5V
FIELD_GND
FIELD_5V_PGOOD
FIELD_GND

STARTUP: R1/R2 bias F_PGOOD from protected input.
Entry TPS26630 PGOOD sinks this node until ready.
REMOVE entry R29; no parallel 5V pull−up or DNP path.
J1.3 is an analog power−enable node, not GPIO.

PGOOD below valid VIN is UNKNOWN.
FIELD_5V_PGOOD is diagnostic only.
R4 is a weak bleed; no rapid discharge guarantee.

Release blockers: input/output capacitor derating; fixed−5V PFM accuracy; custom U1/L1 footprints; entry integration; load/inrush; EMC; layout/thermal/bench tests.
Field output loads MUST remain held off through entry ramp and rail startup; driver/inhibit circuit still required. This supply grants no machine permission.
Boundary connectors are schematic interfaces only. No PCB, DRC, Gerbers, assembly placement, isolation rating or physical qualification is claimed.

Sheet: /field_power/
File: field_power.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 5/14

TWO CURRENT–LIMITED 5V ENCODER SUPPLIES – ENGINEERING BLOCK r0.1

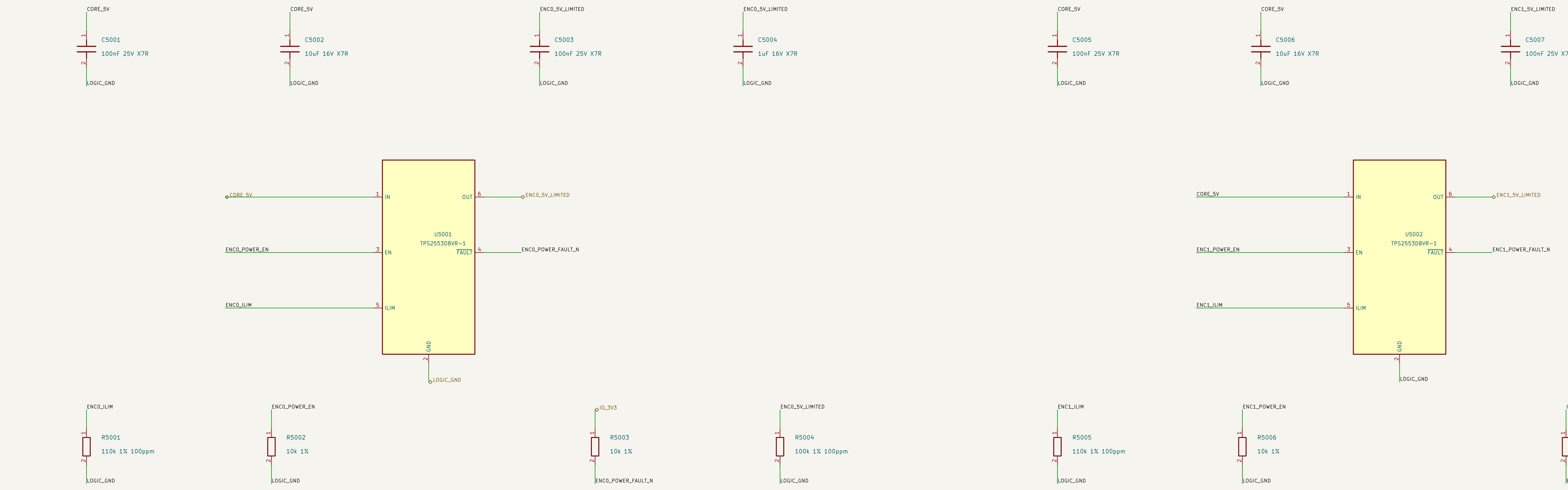
TPS2553DBVR–1: DBV/SOT–23–6, not DRV. Exact receiver net names retained. No main supply/input protection, controller or board layout.

ENCODER 0: INDEPENDENT LATCH–OFF CURRENT LIMIT

CORE_5V required 4.75-5.25V; this switch does not regulate voltage.
200mA encoder load target; output at load depends on upstream and cable drop.

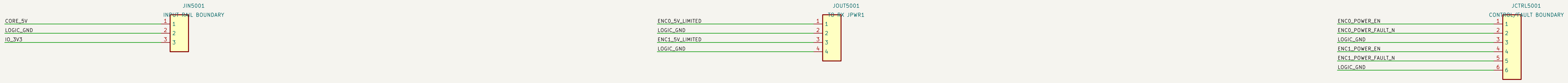
ENCODER 1: INDEPENDENT LATCH–OFF CURRENT LIMIT

CORE_5V required 4.75-5.25V; this switch does not regulate voltage.
200mA encoder load target; output at load depends on upstream and cable drop.


CORE_5V
C5001
100nF 25V X7R
LOGIC_GND
CORE_5V
C5002
10uF 16V X7R
LOGIC_GND
ENC0_5V_LIMITED
C5003
100nF 25V X7R
LOGIC_GND
ENC0_5V_LIMITED
C5004
1uF 16V X7R
LOGIC_GND
CORE_5V
C5005
100nF 25V X7R
LOGIC_GND
CORE_5V
C5006
10uF 16V X7R
LOGIC_GND
ENC1_5V_LIMITED
C5007
100nF 25V X7R
LOGIC_GND
ENC1_5V_LIMITED
C5008
1uF 16V X7R
LOGIC_GND
CORE_5V
ENC0_POWER_EN
ENC0_ILIM
IN
EN
ILIM
U5001
TPS2553DBVR–1
OUT
FAULT
GND
ENC0_5V_LIMITED
ENC0_POWER_FAULT_N
LOGIC_GND
CORE_5V
ENC1_POWER_EN
ENC1_ILIM
IN
EN
ILIM
U5002
TPS2553DBVR–1
OUT
FAULT
GND
ENC1_5V_LIMITED
ENC1_POWER_FAULT_N
LOGIC_GND
ENC0_ILIM
R5001
110k 1% 100ppm
LOGIC_GND
ENC0_POWER_EN
R5002
10k 1%
LOGIC_GND
IO_3V3
R5003
10k 1%
ENC0_POWER_FAULT_N
ENC0_5V_LIMITED
R5004
100k 1% 100ppm
LOGIC_GND
ENC1_ILIM
R5005
110k 1% 100ppm
LOGIC_GND
ENC1_POWER_EN
R5006
10k 1%
LOGIC_GND
IO_3V3
R5007
10k 1%
ENC1_POWER_FAULT_N
ENC1_5V_LIMITED
R5008
100k 1% 100ppm
LOGIC_GND

Place C1 at U1 IN/GND and R1 close to ILIM.
10k EN pulldown = off for a high–Z controller; no hardware restart latch drawn.
FAULT is open–drain, pulled only to IO_3V3; not an output–voltage/power–good measurement.

Place C5 at U2 IN/GND and R5 close to ILIM.
10k EN pulldown = off for a high–Z controller; no hardware restart latch drawn.
FAULT is open–drain, pulled only to IO_3V3; not an output–voltage/power–good measurement.


JIN5001
INPUT RAIL BOUNDARY
CORE_5V
LOGIC_GND
IO_3V3
JOUT5001
TO RX JPWR1
ENC0_5V_LIMITED
LOGIC_GND
ENC1_5V_LIMITED
LOGIC_GND
JCTRL5001
CONTROL/FAULT BOUNDARY
ENC0_POWER_EN
ENC0_POWER_FAULT_N
LOGIC_GND
ENC1_POWER_EN
ENC1_POWER_FAULT_N
LOGIC_GND

110k RILIM including 1%+150K/100ppm allowance: 207.445-283.605mA.
Two simultaneous pre–latch limits: 0.567210A plus switch supply current.
Ledger 0.4A is normal encoder load, not the fault budget. No full power budget approval.

Overcurrent/reverse events latch off after deglitch; EN toggle or power cycle clears latch.
THERMAL SHUTDOWN CAN AUTO–RETRY, including the –1 part. External fault–inhibit policy is missing.
A high EN retained across power return can energize the port. No automatic–restart prevention is claimed.

Not a universal reverse–current barrier: reverse current can flow before protection responds.
No guarantee for 24V/reversed wiring, surge or system ESD. Board/cable faults need qualification.
100k output bleed is passive; no active discharge or guaranteed de–energization time.

EN and FAULT interfaces target 0-3.6V, shared ground; partial–power/backfeed tests remain open.
Do not infer healthy power from an unasserted FAULT, especially with the switch/host unpowered.
Capacitor MPN/effective capacitance and startup/load tests are pending. Unreleased engineering source.

Engineering block; fault interface/physical qualification incomplete

Sheet: /encoder_power/
File: encoder_power.kicad_sch

Title: Two protected encoder supply branches

Size: A2
Date: 2026–09–28
Rev: r0.1
KiCad E.D.A. 10.0.4
Id: 6/14

TWELVE–MOTOR STEP/DIR TRANSMITTER CANDIDATE

24 differential signals / 6 AM26C31IPWR banks. New electrical scope; current firmware remains six–channel.

STEP0_FPGA
DIR0_FPGA
STEP1_FPGA
DIR1_FPGA
R6001
R6002
R6003
R6004
33R 1%
CORE_5V
C6001
100nF 25V X7R
C6002
1uF 25V X7R
STEP0_D
DIR0_D
STEP1_D
DIR1_D
R6025
R6026
R6027
R6028
10k 1%
LOGIC_GND
TX_PERMIT
R6049
U6001
AM26C31IPWR
VCC
GND
G
1A
2A
3A
4A
1Y
1Z
2Y
2Z
3Y
3Z
4Y
4Z
STEP0_P
STEP0_N
DIR0_P
DIR0_N
STEP1_P
STEP1_N
DIR1_P
DIR1_N
JTX6000
FIELD TERMINALS / MPN TBD
R6005
R6006
R6007
R6008
C6003
C6004
STEP2_FPGA
DIR2_FPGA
STEP3_FPGA
DIR3_FPGA
R6029
R6030
R6031
R6032
R6050
U6002
STEP2_P
STEP2_N
DIR2_P
DIR2_N
STEP3_P
STEP3_N
DIR3_P
DIR3_N
JTX6001
R6009
R6010
R6011
R6012
C6005
C6006
STEP4_FPGA
DIR4_FPGA
STEP5_FPGA
DIR5_FPGA
R6033
R6034
R6035
R6036
R6051
U6003
STEP4_P
STEP4_N
DIR4_P
DIR4_N
STEP5_P
STEP5_N
DIR5_P
DIR5_N
JTX6002
JLOGIC6001
INTERNAL LOGIC BOUNDARY
1 CORE_5V
2 LOGIC_GND
3 TX_PERMIT
4 STEP0_FPGA
5 DIR0_FPGA
6 STEP1_FPGA
7 DIR1_FPGA
8 STEP2_FPGA
9 DIR2_FPGA
10 STEP3_FPGA
11 DIR3_FPGA
12 STEP4_FPGA
13 DIR4_FPGA
14 STEP5_FPGA
15 DIR5_FPGA
16 STEP6_FPGA
17 DIR6_FPGA
18 STEP7_FPGA
19 DIR7_FPGA
20 STEP8_FPGA
21 DIR8_FPGA
22 STEP9_FPGA
23 DIR9_FPGA
24 STEP10_FPGA
25 DIR10_FPGA
26 STEP11_FPGA
27 DIR11_FPGA
28 LOGIC_GND
JREF6001
SIGNAL RETURNS
1 LOGIC_GND
2 LOGIC_GND
3 LOGIC_GND
4 LOGIC_GND
5 LOGIC_GND
6 LOGIC_GND
R6013
R6014
R6015
R6016
C6007
C6008
STEP6_FPGA
DIR6_FPGA
STEP7_FPGA
DIR7_FPGA
R6037
R6038
R6039
R6040
R6052
U6004
STEP6_P
STEP6_N
DIR6_P
DIR6_N
STEP7_P
STEP7_N
DIR7_P
DIR7_N
JTX6003
R6017
R6018
R6019
R6020
C6009
C6010
STEP8_FPGA
DIR8_FPGA
STEP9_FPGA
DIR9_FPGA
R6041
R6042
R6043
R6044
R6053
U6005
STEP8_P
STEP8_N
DIR8_P
DIR8_N
STEP9_P
STEP9_N
DIR9_P
DIR9_N
JTX6004
R6021
R6022
R6023
R6024
C6011
C6012
STEP10_FPGA
DIR10_FPGA
STEP11_FPGA
DIR11_FPGA
R6045
R6046
R6047
R6048
R6054
U6006
STEP10_P
STEP10_N
DIR10_P
DIR10_N
STEP11_P
STEP11_N
DIR11_P
DIR11_N
JTX6005

BANK TX0 / U1: JOINTS 0 AND 1

Place C1 close to U1 pins 16/8.
JTX0 matches allocation positions 1-8; reference is separate.

BANK TX1 / U2: JOINTS 2 AND 3

Place C3 close to U2 pins 16/8.
JTX1 matches allocation positions 1-8; reference is separate.

BANK TX2 / U3: JOINTS 4 AND 5

Place C5 close to U3 pins 16/8.
JTX2 matches allocation positions 1-8; reference is separate.

Engineering circuit only.
All twelve channels populated.
8–channel population needs
separate BOM and export.
No price or speed approval.

BANK TX3 / U4: JOINTS 6 AND 7

Place C7 close to U4 pins 16/8.
JTX3 matches allocation positions 1-8; reference is separate.

BANK TX4 / U5: JOINTS 8 AND 9

Place C9 close to U5 pins 16/8.
JTX4 matches allocation positions 1-8; reference is separate.

BANK TX5 / U6: JOINTS 10 AND 11

Place C11 close to U6 pins 16/8.
JTX5 matches allocation positions 1-8; reference is separate.

JLOGIC1 is an internal boundary.
3.3V logic; no 5V GPIO pullups.
No inputs driven with VCC off.
Six enable pulldowns in parallel.
TX_PERMIT requires >=2V high.
No line protection or PCB yet.
No automatic receiver fail state.
Drive enables/alarms separate.

TX_PERMIT low makes differential outputs HIGH–Z. A remote drive may interpret floating lines incorrectly; receiver bias and independent drive inhibit need qualification.
1.2A reservation on CORE_5V is a conservative design allowance, not a measured load. Existing six–channel converter budget cannot be reused without recalculation.
Twelve motor channels are not twelve Cartesian axes. No gateware expansion, cable qualification, thermal proof or manufacturing release follows from this drawing.

Source: TI AM26C31 SLLS103P, pin Table4–1, operating Section5.3, electrical Section5.5, function Table7–1.
Input33R damping,10k defaults, capacitor MPNs, connector selections and final protection remain open.

Partial schematic only; not released for manufacture

Sheet: /tx_twelve/
File: tx_twelve.kicad_sch

Title: Twelve–channel differential transmitter candidate

Size: A1	Date: 2026–09–28	**Rev: r0.2**
KiCad E.D.A. 10.0.4		Id: 7/14

BASELINE TWO–ABZ DIFFERENTIAL RECEIVER – SCHEMATIC BLOCK r0.1

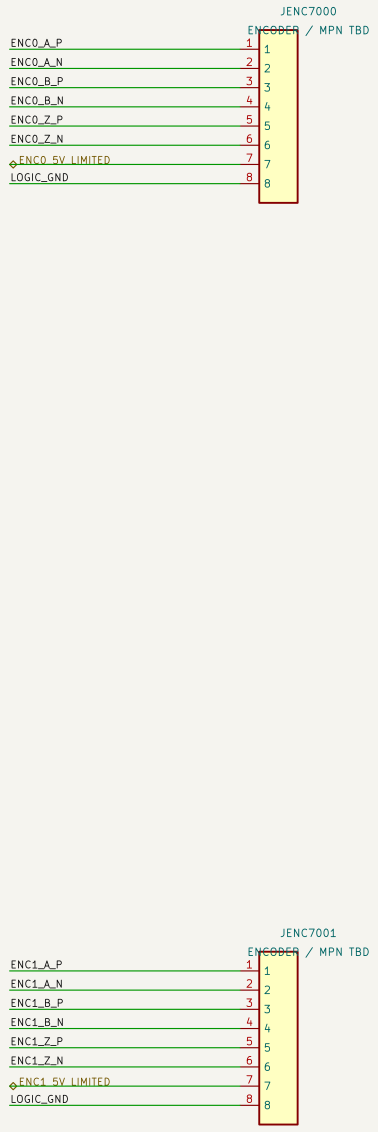
Six receiver channels, exact ledger allocation; no encoder supply limiter, line protection, FPGA pin assignment or board release.

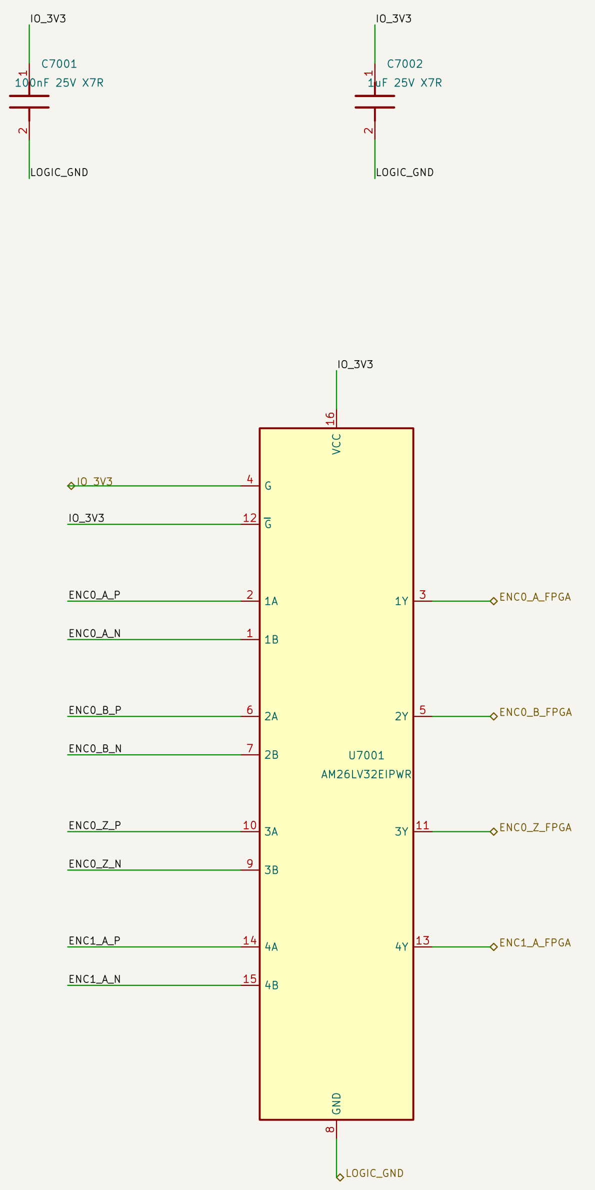
U1 = LEDGER RX0

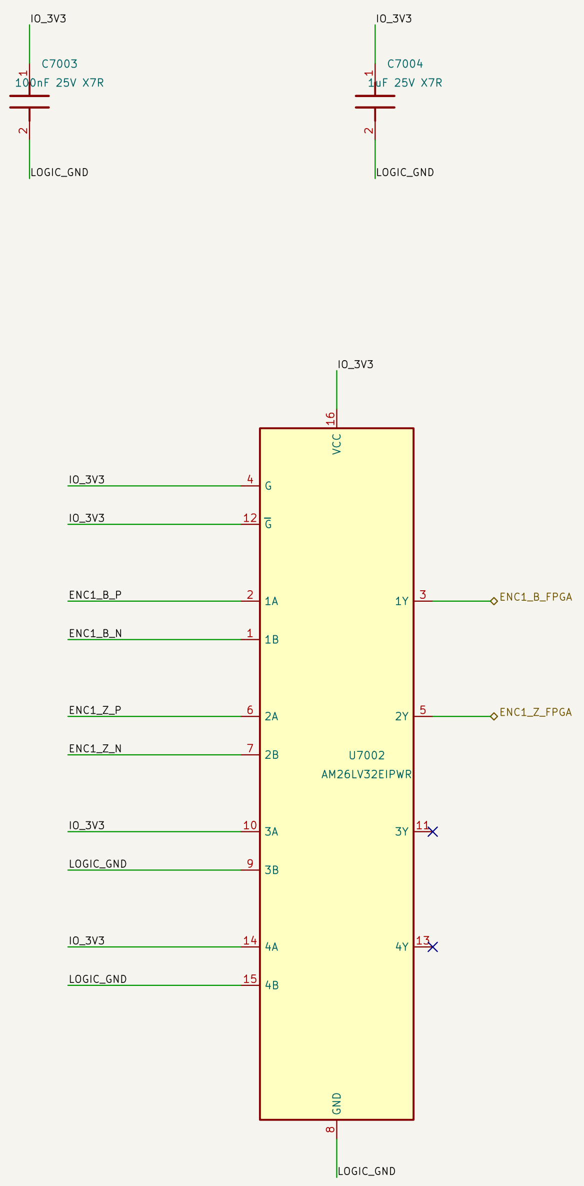
AM26LV32EIPWR / PW16 / 3.0-3.6V supply
G=/G=3.3V: always enabled; receiver state does not grant motion.

U2 = LEDGER RX1

AM26LV32EIPWR / PW16 / 3.0-3.6V supply
G=/G=3.3V: always enabled; receiver state does not grant motion.

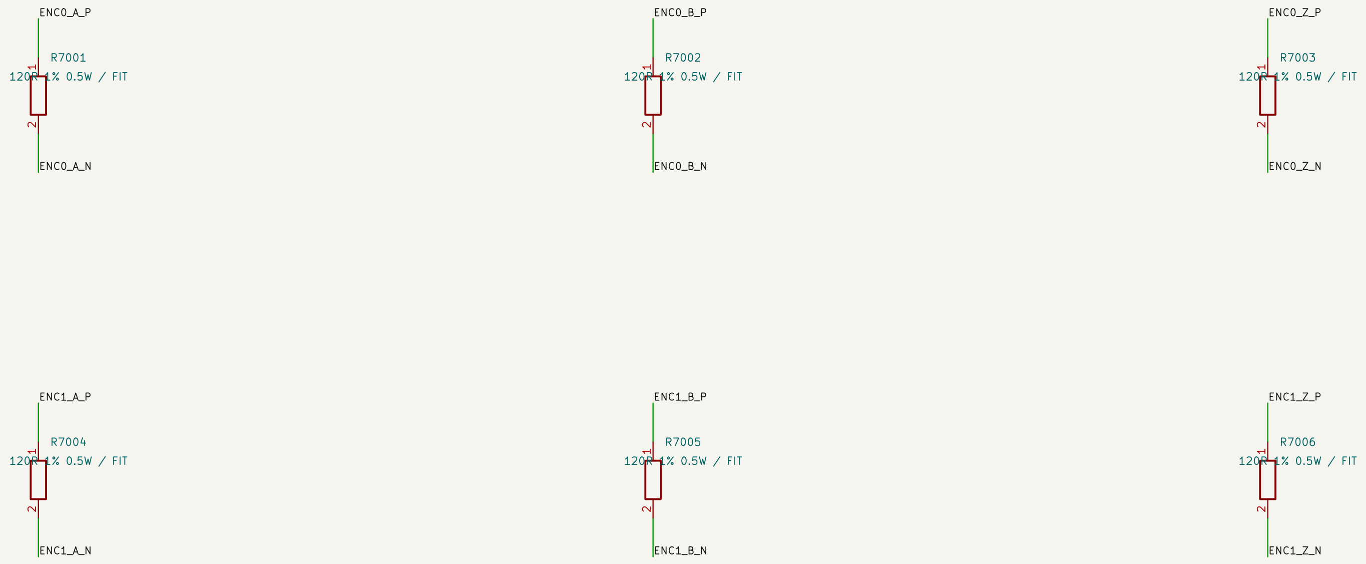

JENC7000
ENCODER / MPN TBD
ENC0_A_P
ENC0_A_N
ENC0_B_P
ENC0_B_N
ENC0_Z_P
ENC0_Z_N
ENC0_5V_LIMITED
LOGIC_GND
1
2
3
4
5
6
7
8
JENC7001
ENCODER / MPN TBD
ENC1_A_P
ENC1_A_N
ENC1_B_P
ENC1_B_N
ENC1_Z_P
ENC1_Z_N
ENC1_5V_LIMITED
LOGIC_GND


IO_3V3
C7001
100nF 25V X7R
LOGIC_GND
IO_3V3
C7002
1uF 25V X7R
LOGIC_GND
IO_3V3
16
VCC
IO_3V3
4
G
IO_3V3
12
G̅
ENC0_A_P
2
1A
ENC0_A_N
1
1B
ENC0_B_P
6
2A
ENC0_B_N
7
2B
ENC0_Z_P
10
3A
ENC0_Z_N
9
3B
ENC1_A_P
14
4A
ENC1_A_N
15
4B
U7001
AM26LV32EIPWR
1Y
3
ENC0_A_FPGA
2Y
5
ENC0_B_FPGA
3Y
11
ENC0_Z_FPGA
4Y
13
ENC1_A_FPGA
GND
8
LOGIC_GND

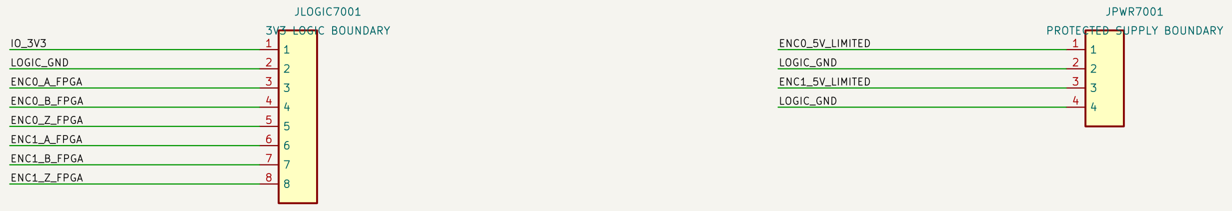

IO_3V3
C7003
100nF 25V X7R
LOGIC_GND
IO_3V3
C7004
1uF 25V X7R
LOGIC_GND
IO_3V3
16
VCC
IO_3V3
4
G
IO_3V3
12
G̅
ENC1_B_P
2
1A
ENC1_B_N
1
1B
ENC1_Z_P
6
2A
ENC1_Z_N
7
2B
IO_3V3
10
3A
LOGIC_GND
9
3B
IO_3V3
14
4A
LOGIC_GND
15
4B
U7002
AM26LV32EIPWR
1Y
3
ENC1_B_FPGA
2Y
5
ENC1_Z_FPGA
3Y
11
4Y
13
GND
8
LOGIC_GND

Unused U2 channels 3/4:
A tied IO_3V3; B tied LOGIC_GND.
Outputs 11/13 deliberately NC.
No floating or shorted unused inputs.

TERMINATION: fitted 120R per pair; DNP = open. Never substitute a short.
Location is receiver end only; encoder/cable profile must qualify the load.


ENC0_A_P
R7001
120R 1% 0.5W / FIT
ENC0_A_N
ENC0_B_P
R7002
120R 1% 0.5W / FIT
ENC0_B_N
ENC0_Z_P
R7003
120R 1% 0.5W / FIT
ENC0_Z_N
ENC1_A_P
R7004
120R 1% 0.5W / FIT
ENC1_A_N
ENC1_B_P
R7005
120R 1% 0.5W / FIT
ENC1_B_N
ENC1_Z_P
R7006
120R 1% 0.5W / FIT
ENC1_Z_N
1
2

JENC pin1/2=A+/–; 3/4=B+/–; 5/6=Z+/–; 7=protected5V; 8=logic ground.
Supply limiters are absent: JPWR1 passes separately supplied, already protected rails.
JLOGIC1/JPWR1 are circuit boundaries only; connector MPNs remain unselected.


JLOGIC7001
3V3 LOGIC BOUNDARY
IO_3V3
LOGIC_GND
ENC0_A_FPGA
ENC0_B_FPGA
ENC0_Z_FPGA
ENC1_A_FPGA
ENC1_B_FPGA
ENC1_Z_FPGA
1
2
3
4
5
6
7
8
JPWR7001
PROTECTED SUPPLY BOUNDARY
ENC0_5V_LIMITED
LOGIC_GND
ENC1_5V_LIMITED
LOGIC_GND
1
2
3
4

Termination reference: CRCW1210120RFKEA (1210,0.5W at70C).
Declared applied differential bound:5.5V; worst allowance ~0.259W/channel.
Initial localambient<=70C; PCB/film temperature and load still require measurement.
Receiver +/–7V common mode does not rate termination for14V differential.
ESD component ratings do not qualify the complete controller.

TI SLLS849E: Fig5–1 / Table8–1 / Section10. Place C1/C3 close to U1/U2 pins16/8.
Open–circuit fail–safe does NOT guarantee a valid state for a shorted or terminated disconnected cable.
No galvanic isolation, independent cable–loss detection or measured encoder timing here.

Partial schematic; no physical qualification or manufacturing release

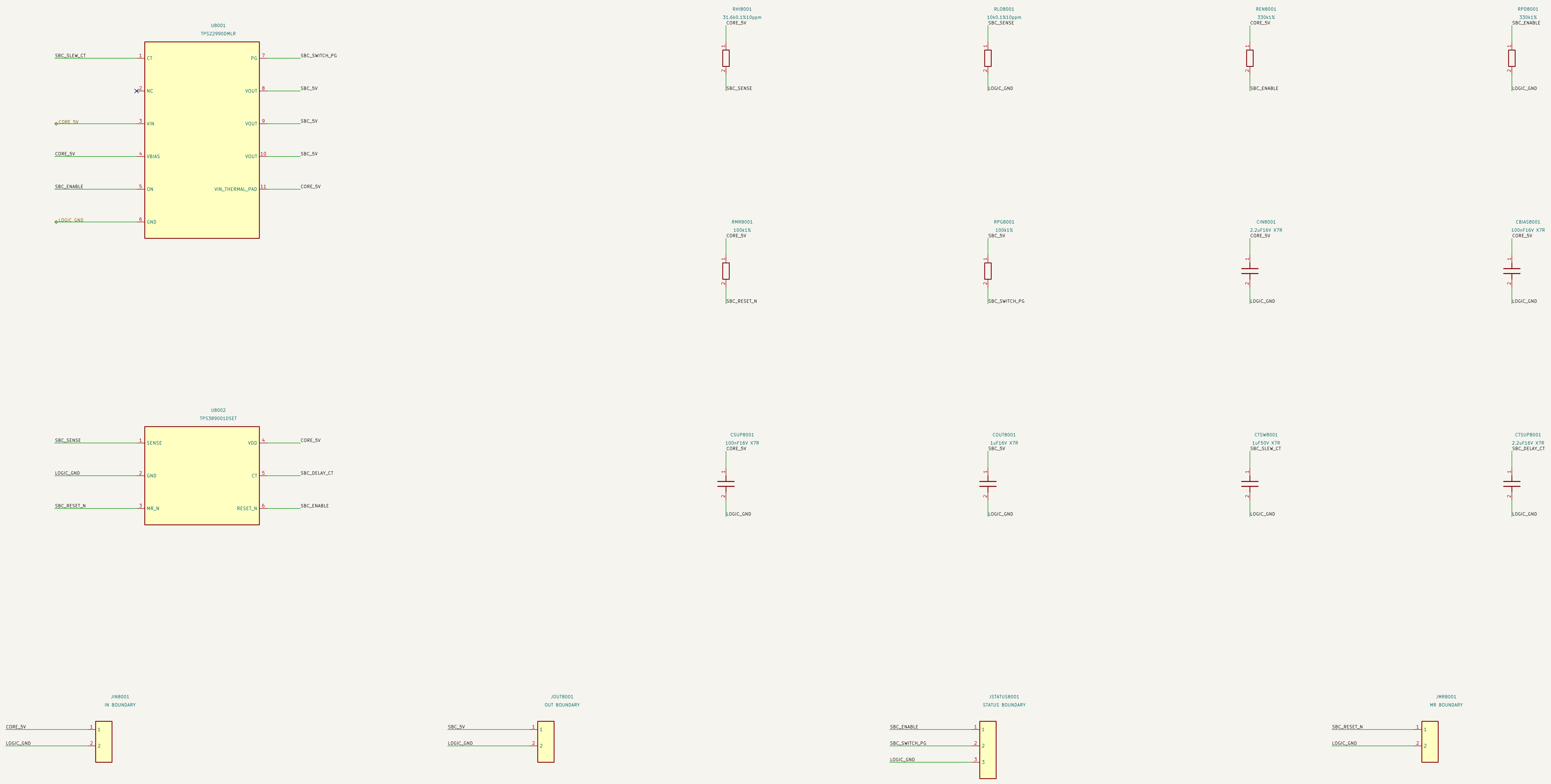
Sheet: /rx_abz/
File: rx_abz.kicad_sch

Title: Baseline two–ABZ differential receiver

Size: A2
Date: 2026–09–28
Rev: r0.1
KiCad E.D.A. 10.0.4
Id: 8/14

HARDWARE–SUPERVISED SBC POWER – 3 A BRANCH CANDIDATE

CORE_5V charges first. Supervisor release and controlled switch slew start SBC_5V without host software.
U1 pad11 is VIN, never GND. U2 has six terminals and no exposed thermal pad. All footprints remain unassigned.


U8001
TPS22990DMLR
SBC_SLEW_CT 1 CT
2 NC
CORE_5V 3 VIN
CORE_5V 4 VBIAS
SBC_ENABLE 5 ON
LOGIC_GND 6 GND
PG 7 SBC_SWITCH_PG
VOUT 8 SBC_5V
VOUT 9 SBC_5V
VOUT 10 SBC_5V
VIN_THERMAL_PAD 11 CORE_5V
U8002
TPS389001DSET
SBC_SENSE 1 SENSE
LOGIC_GND 2 GND
SBC_RESET_N 3 MR_N
VDD 4 CORE_5V
CT 5 SBC_DELAY_CT
RESET_N 6 SBC_ENABLE
RHI8001
31.6k0.1%10ppm
CORE_5V
SBC_SENSE
RLO8001
10k0.1%10ppm
SBC_SENSE
LOGIC_GND
REN8001
330k1%
CORE_5V
SBC_ENABLE
RPD8001
330k1%
SBC_ENABLE
LOGIC_GND
RMR8001
100k1%
CORE_5V
SBC_RESET_N
RPG8001
100k1%
SBC_5V
SBC_SWITCH_PG
CIN8001
2.2uF16V X7R
CORE_5V
LOGIC_GND
CBIAS8001
100nF16V X7R
CORE_5V
LOGIC_GND
CSUP8001
100nF16V X7R
CORE_5V
LOGIC_GND
COUT8001
1uF16V X7R
SBC_5V
LOGIC_GND
CTSW8001
1uF50V X7R
SBC_SLEW_CT
LOGIC_GND
CTSUP8001
2.2uF16V X7R
SBC_DELAY_CT
LOGIC_GND
JIN8001
IN BOUNDARY
CORE_5V 1
LOGIC_GND 2
JOUT8001
OUT BOUNDARY
SBC_5V 1
LOGIC_GND 2
JSTATUS8001
STATUS BOUNDARY
SBC_ENABLE 1
SBC_SWITCH_PG 2
LOGIC_GND 3
JMR8001
MR BOUNDARY
SBC_RESET_N 1
LOGIC_GND 2

JMR1 accepts an optional normally–open dry contact to ground. No external power or active–high driver.
JSTATUS1 is a 5 V–domain diagnostic boundary, not direct 3.3 V GPIO. Switch PG means enhanced FET, not valid output voltage.

No branch current limiting, reverse blocking or arbitrary–brownout hold–up is supplied by this block.
CTSW1 requires >=470 nF effective; CTSUP1 >=1.2 uF effective. Values and timing require qualified parts/tests.
SBC total charge capacitance <=1 mF; host current <=3 A. No alternative USB/SBC power source until backfeed is reviewed.
No machine permission follows from power return. Circuit ERC is not PCB or physical qualification.

Sheet: /sbc_power/
File: sbc_power.kicad_sch
Title: Sequenced SBC supply
Size: A2 Date: 2026–09–28 **Rev: r0.1**
KiCad E.D.A. 10.0.4 Id: 9/14

AUTONOMOUS POWER WINDOW / RESET CANDIDATE — NOT MACHINE PERMISSION

R9001
87.9k 0.01% 5ppm
R9009
92.1k 0.01% 5ppm
C9001
100nF 25V X7R 20%
R9007
10k 0.01% 5ppm
R9015
10k 0.01% 5ppm
U9001
LTC2962HUD#TRPBF
VCC
V1
V2
V3
V4
MR
RDIS
DVCC
RST
REF
PG1
PG2
PG3
PG4
RT
GND
EP
CORE_5V
CORE_5V_UV_SENSE
CORE_5V_OV_SENSE
LOGIC_GND
SEQ_5V_OK
SEQ_REF1
R9101
3.3k 1%
R9102
3.3k 1%
R9104
3.3k 1%
R9103
10k 1%
R9105
10k 1%
CORE_1V2_EN
ESP_3V3_EN
R9021
1.10k 0.01% 5ppm
R9029
1.15k 0.01% 5ppm
C9002
100nF 25V X7R 20%
R9022
255 0.01% 5ppm
R9030
294 0.01% 5ppm
R9031
1 1% 100ppm
R9027
1k 0.01% 5ppm
R9035
1k 0.01% 5ppm
CORE_1V2
SEQ_UV2_1
SEQ_OV2_1
SEQ_OV2_2
CORE_1V2_UV_SENSE
CORE_1V2_OV_SENSE
U9002
LTC2962HUD#TRPBF
IO_3V3_EN
SEQ_REF2
R9106
6.8k 1%
R9107
10k 1%
R9041
54.5k 0.01% 5ppm
R9049
57.5k 0.01% 5ppm
C9003
100nF 25V X7R 20%
R9047
10k 0.01% 5ppm
R9055
10k 0.01% 5ppm
IO_3V3
IO_3V3_UV_SENSE
IO_3V3_OV_SENSE
U9003
LTC2962HUD#TRPBF
SEQ_FPGA_RESET_GOOD
SEQ_REF3
R9108
47k 1%
R9109
470k 1%
U9005
SN74LVC1G17DBVR
NC
A
FPGA_CRESET_N
C9005
100nF 25V X7R 20%
R9110
4.7k 1%
R9061
54.5k 0.01% 5ppm
R9069
57.5k 0.01% 5ppm
C9004
100nF 25V X7R 20%
R9067
10k 0.01% 5ppm
R9075
10k 0.01% 5ppm
ESP_3V3
ESP_3V3_UV_SENSE
ESP_3V3_OV_SENSE
U9004
LTC2962HUD#TRPBF
SEQ_ESP_RESET_GOOD
SEQ_REF4
R9111
47k 1%
R9112
470k 1%
U9006
SN74LVC1G17DBVR
ESP_CHIP_PU
C9006
100nF 25V X7R 20%
R9113
47k 1%
JRAIL9001
INTEGRATION_BOUNDARY
JCTRL9001
INTEGRATION_BOUNDARY
#FLG9004
EXTERNAL

1. CORE_5V window / autonomous release

PG1=REF: UV; PG2=GND: OV. Unused V3/V4=VCC, PG3/4=REF.
RT=VCC: 160–240ms release. RDIS=VCC: margin override disabled.

2. CORE_1V2 window / autonomous release

PG1=REF: UV; PG2=GND: OV. Unused V3/V4=VCC, PG3/4=REF.
RT=VCC: 160–240ms release. RDIS=VCC: margin override disabled.

3. IO_3V3 window / autonomous release

PG1=REF: UV; PG2=GND: OV. Unused V3/V4=VCC, PG3/4=REF.
RT=VCC: 160–240ms release. RDIS=VCC: margin override disabled.

4. ESP_3V3 window / autonomous release

PG1=REF: UV; PG2=GND: OV. Unused V3/V4=VCC, PG3/4=REF.
RT=VCC: 160–240ms release. RDIS=VCC: margin override disabled.

CORE5 –> core + ESP enables; valid core –> IO enable. IO qualifier –> FPGA reset.
ESP qualifier –> ESP reset only: ESP fault does not reset FPGA through this circuit.
Every EN also has the existing regulator10k pulldown. No active–high external driver allowed.
5V/internal pullups never directly drive destination reset pins. Schmitt buffers use destination rails.

Core threshold static margins are sub–mV at some corners. Aging, stress, ripple and transient
response remain release blockers. Comparator40us maximum applies only at10% overdrive.
No guaranteed brownout slew, powered–off regulator discharge, shutdown order or restart latch.
FPGA package/PLL filters, configuration, CDONE/user–mode guard and machine inhibit remain external.
Four precision supervisors are a COST–RISK CANDIDATE, not a production BOM choice.

Engineering candidate; no PCB or physical validation

Sheet: /power_sequence/
File: power_sequence.kicad_sch

Title: Autonomous rail qualification and reset

Size: A1 | Date: 2026–09–28 | **Rev: r0.1**
KiCad E.D.A. 10.0.4 | Id: 10/14

BASELINE 22 ISOLATED SINKING DIGITAL INPUTS — 6 HOME + 16 AUXILIARY — REVIEW r0.1

0–30V input study; low<=5V / high>=15V targets. PNP source or dry contact to field positive. Common FIELD_GND; no per–channel isolation.
11 ISO1212 devices require IO_3V3 only. Do not join FIELD_GND to LOGIC_GND. SUB pins get separate floating copper islands; never ground or join them.

DI0: DI00 / DI01 — FIELD inputs –> LOGIC
DI1: DI02 / DI03 — FIELD inputs –> LOGIC
DI2: DI04 / DI05 — FIELD inputs –> LOGIC
DI3: DI06 / DI07 — FIELD inputs –> LOGIC
DI4: DI08 / DI09 — FIELD inputs –> LOGIC
DI5: DI10 / DI11 — FIELD inputs –> LOGIC
DI6: DI12 / DI13 — FIELD inputs –> LOGIC
DI7: DI14 / DI15 — FIELD inputs –> LOGIC
DI8: DI16 / DI17 — FIELD inputs –> LOGIC
DI9: DI18 / DI19 — FIELD inputs –> LOGIC
DI10: DI20 / DI21 — FIELD inputs –> LOGIC
U10001
ISO1212DBQR
SENSE1
IN1
FGND1
SUB1
SENSE2
IN2
FGND2
SUB2
VCC1
EN
OUT1
OUT2
GND1
NC
DI00_SENSE
DI00_IN
FIELD_GND
DI00_SUB_ISLAND
DI01_SENSE
DI01_IN
DI01_SUB_ISLAND
IO_3V3
HOME0_FPGA
HOME1_FPGA
LOGIC_GND
C10001
10nF 100V C0G
C10101
100nF 25V X7R
R10201
100k 1%
R10202
100k 1%
R10018
1k MELF 1%
R10118
562R 1%
C10018
10nF 100V C0G
DI17_FIELD
DI17_SENSE
DI17_IN
JDI10000
JDI0 / LEDGER
DI00_FIELD
DI01_FIELD
DI02_FIELD
DI03_FIELD
DI04_FIELD
DI05_FIELD
DI06_FIELD
DI07_FIELD
JDI10001
JDI1 / LEDGER
DI08_FIELD
DI09_FIELD
DI10_FIELD
DI11_FIELD
DI12_FIELD
DI13_FIELD
DI14_FIELD
DI15_FIELD
JDI10002
JDI2 / LEDGER
DI16_FIELD
DI17_FIELD
DI18_FIELD
DI19_FIELD
DI20_FIELD
DI21_FIELD
FIELD_GND
FIELD_GND
JLOG10001
INTERNAL LOGIC BOUNDARY
IO_3V3
LOGIC_GND
HOME0_FPGA
HOME1_FPGA
HOME2_FPGA
HOME3_FPGA
HOME4_FPGA
HOME5_FPGA
PROBE_FPGA
TOOLSETTER_FPGA
DOOR_STATUS_FPGA
PROTECTIVE_STATUS_FPGA
SPINDLE_FAULT_FPGA
SPINDLE_READY_FPGA
JLOG10002
INTERNAL LOGIC BOUNDARY
IO_3V3
LOGIC_GND
FEED_HOLD_FPGA
CYCLE_START_FPGA
LUBE_STATUS_FPGA
SPARE_FPGA
ALM0_FPGA
ALM1_FPGA
ALM2_FPGA
ALM3_FPGA
ALM4_FPGA
ALM5_FPGA
R10219
100k 1%
R10220
100k 1%

All JDI names/positions match frozen ledger.
JDI0/1 have no local common terminal: use JDI2.7/.8.
No sensor positive supply is provided by this sheet.
Boundary symbols are not approved connector MPNs.
No PE capacitor fitted: whole–system coupling/EMC is open.

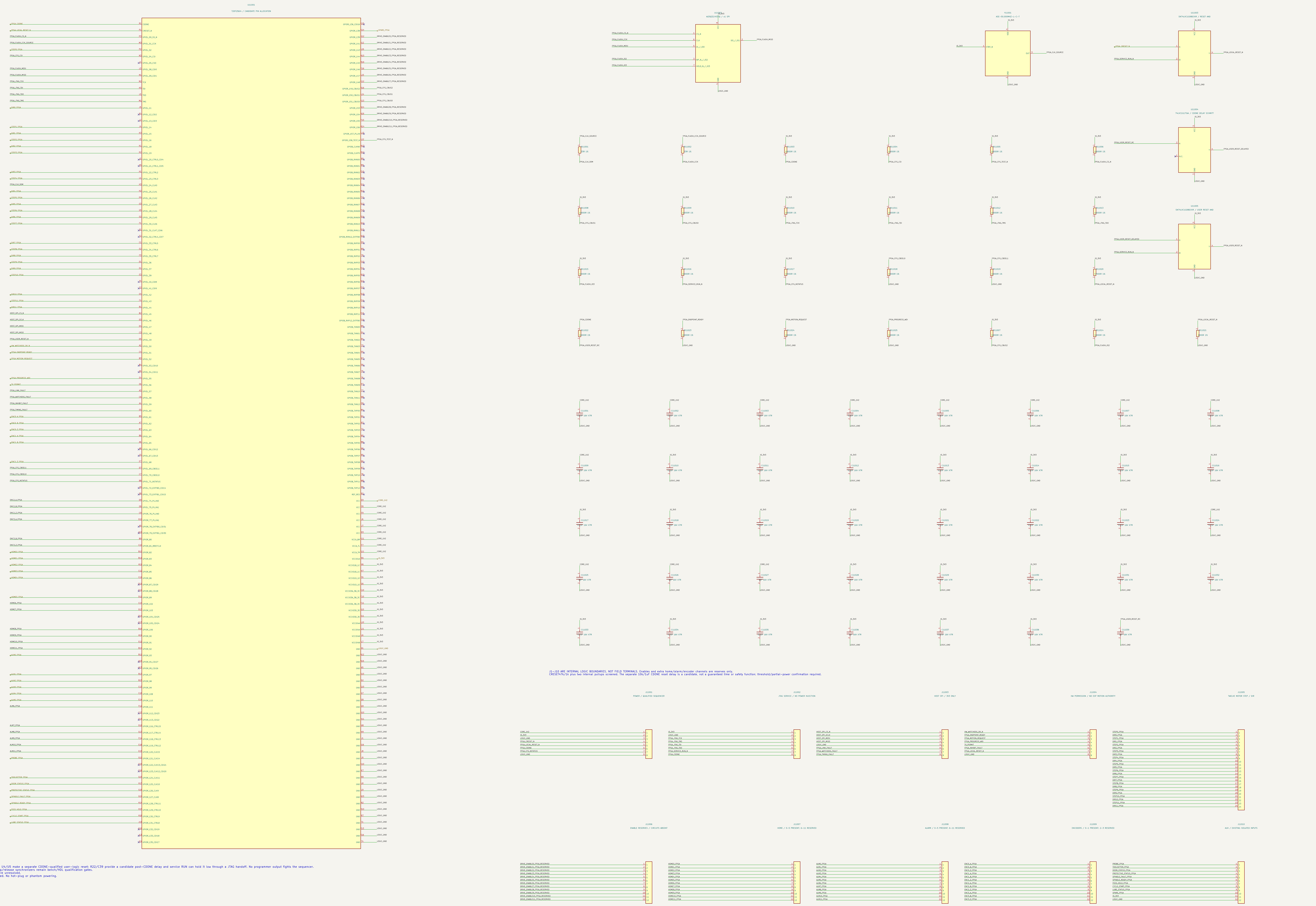
EN is tied to IO_3V3. Powered open input reads LOW. Unpowered/brownout output is UNDETERMINED; pull–down is bias, not a fail–safe guarantee.
Mask all inputs until qualified rail/FPGA readiness. PROBE/TOOLSETTER filters need latency qualification. No safety input certification, physical EMC result or PCB release.

Sheet: /digital_inputs/
File: digital_inputs.kicad_sch
Title:
Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 11/14

T20F256 FPGA CORE / MINIMUM 8 MOTOR CHANNELS, 12–CHANNEL PIN CAPACITY CANDIDATE

Exact 256 balls: Efinix T20–PINOUT worksheet T20F256. Pin allocation is a board candidate, NOT vendor fit or routed timing.
All VCCIO=3.3V. All VCC/VCCA=1.2V. No 5V–tolerant FPGA connection. No native LVDS: REF_RES intentionally NC.

One 100nF per supply ball; bulk per physical supply group. Decoupling is a layout–study allowance, not a PDN proof.
Active SPI x1 straps111; CSI/TEST high. Flash bus reserved from user RTL. CCK damping33R is provisional.
CDONE is open–drain with10k pullup; status does not grant motion permission. Configuration clock is internal, not Y1.



U11001
T20F256I4 / CANDIDATE PIN ALLOCATION
U11002
W25Q32JVSSIQ / x1 SPI
Y11001
ASE–50.000MHZ–L–C–T
FPGA_CLK_SOURCE
U11003
SN74LVC1G08DCKR / RESET AND
FPGA_CRESET_N
FPGA_SERVICE_RUN_N
FPGA_LOCAL_RESET_N
U11004
74LVC1G17GW / CDONE DELAY SCHMITT
FPGA_USER_RESET_DELAYED
IO_3V3
LOGIC_GND
CORE_1V2
J11001
POWER / QUALIFIED SEQUENCER
J11002
JTAG SERVICE / NO POWER INJECTION
J11003
HOST SPI / 3V3 ONLY
J11004
HW PERMISSION / NO ESP MOTION AUTHORITY
J11005
TWELVE MOTOR STEP / DIR
J11006
ENABLE RESERVES / CIRCUITS ABSENT
J11007
HOME / 0–5 PRESENT; 6–11 RESERVED
J11008
ALARM / 0–5 PRESENT; 6–11 RESERVED
J11009
ENCODERS / 0–1 PRESENT; 2–3 RESERVED
J11010
AUX / EXISTING ISOLATED INPUTS
#FLG11001
SUPPLIED BY UPSTREAM

J1–J10 ARE INTERNAL LOGIC BOUNDARIES, NOT FIELD TERMINALS. Enables and extra home/alarm/encoder channels are reserves only.
CRESET47k/1k plus two internal pullups screened. The separate 10k/1uF CDONE reset delay is a candidate, not a guaranteed time or safety function; threshold/partial–power confirmation required.

RESET: U3 combines qualified upstream reset with service RUN for configuration reset only. U4/U5 make a separate CDONE–qualified user–logic reset; R22/C39 provide a candidate post–CDONE delay and service RUN can hold it low through a JTAG handoff. No programmer output fights the sequencer.
User–logic reset release threshold, RC tolerance, discharge time, tUSER handoff and startup/release synchronizers remain bench/HDL qualification gates.
J2 is a logical service boundary: actual keyed connector, adapter, cable loading and ESD are unresolved.
JTAG programming requires a deliberate reset cycle, valid rails and machine outputs inhibited. No hot–plug or phantom powering.

NC crosses are genuine unused GPIO/config outputs and REF_RES only; every supply/GND ball is connected.
Boundary symbols are not released connectors. No footprint approval, PCB, DRC, bitstream, qualification or fabrication files.

Sheet: /fpga_core/
File: fpga_core.kicad_sch

Title:

Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.4

Id: 12/14

INDEPENDENT WATCHDOG AND EDGE–ARMED GLOBAL PERMISSION – ENGINEERING CANDIDATE

Seven ICs; no firmware timer owns the external watchdog. IO_3V3=3.2–3.4V, common LOGIC_GND. Eight motors minimum / twelve expansion requirement.

1. Always–enabled watchdog; WDO + ENOUT wired together
U12001
TPS3431SDRBR
IO_3V3 1 VDD
WD_CWD 2 CWD
IO_3V3 3 EN
LOGIC_GND 4 GND
IO_3V3 5 SET1
WD_INPUT_BUF 6 WDI
LOGIC_GND 9 EP
WDO_N 7 WD_HEALTH_RAW_N
ENOUT 8 WD_HEALTH_RAW_N
WD_CWD
C12001
120pF C0G 50V 5%
LOGIC_GND
IO_3V3
R12001
10k 1%
WD_HEALTH_RAW_N
2. Schmitt inputs for open–drain watchdog and CDONE
U12002
SN74LVC2G17DBVR
WD_HEALTH_RAW_N 1 1A
LOGIC_GND 2 GND
FPGA_CDONE 3 2A
IO_3V3 5 VCC
2Y 4 CDONE_BUF
1Y 6 WD_HEALTH_BUF
3. Rail–to–rail WDI buffer; FALLING edges feed timer
U12003
SN74LVC1G34DBVR
1 NC
FPGA_PROGRESS_WDI 2 A
LOGIC_GND 3 GND
IO_3V3 5 VCC
Y 4 WD_INPUT_BUF
WD_INPUT_BUF
R12002
100k 1%
LOGIC_GND
FPGA_PROGRESS_WDI
R12003
10k 1%
LOGIC_GND
FPGA_LOCAL_RESET_N
R12004
10k 1%
LOGIC_GND
PROTECTIVE_STATUS_FPGA
R12005
10k 1%
LOGIC_GND
FPGA_ENDPOINT_READY
R12006
10k 1%
LOGIC_GND
FPGA_MOTION_REQUEST
R12007
10k 1%
LOGIC_GND
LATCH_CLEAR_N
R12008
10k 1%
LOGIC_GND
4. Raw interlock
U12004
SN74LVC1G11DBVR
FPGA_LOCAL_RESET_N 1 A
LOGIC_GND 2 GND
CDONE_BUF 3 B
IO_3V3 5 VCC
PROTECTIVE_STATUS_FPGA 6 C
Y 4 BASE_INTERLOCK_OK
5. Raw return and latch clear
U12005
SN74LVC2G08DCUR
BASE_INTERLOCK_OK 1 1A
WD_HEALTH_BUF 2 1B
LOGIC_GND 4 GND
HW_WATCHDOG_OK_N 5 2A
FPGA_ENDPOINT_READY 6 2B
IO_3V3 8 VCC
2Y 3 LATCH_CLEAR_N
1Y 7 HW_WATCHDOG_OK_N
HW_WATCHDOG_OK_N
R12011
10k 1%
LOGIC_GND
6. Accepted ARM edge; async clear
U12006
SN74LVC2G74DCUR
FPGA_MOTION_REQUEST 1 CLK
IO_3V3 2 D
LOGIC_GND 4 GND
LATCH_CLEAR_N 6 CLR_N
IO_3V3 7 PRE_N
IO_3V3 8 VCC
Q_N 3
Q 5 HW_PERMISSION_LATCHED
7. Final physical permit
U12007
SN74LVC1G11DBVR
HW_PERMISSION_LATCHED 1 A
LOGIC_GND 2 GND
FPGA_MOTION_REQUEST 3 B
IO_3V3 5 VCC
HW_WATCHDOG_OK_N 6 C
Y 4 TX_PERMIT
HW_PERMISSION_LATCHED
R12009
10k 1%
LOGIC_GND
IO_3V3
C12002
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12003
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12004
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12005
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12006
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12007
100nF X7R 25V 20%
LOGIC_GND
IO_3V3
C12008
100nF X7R 25V 20%
LOGIC_GND
TX_PERMIT
R12010
10k 1%
LOGIC_GND
JCTRL12001
INTERNAL BOUNDARY
IO_3V3 1
LOGIC_GND 2
FPGA_LOCAL_RESET_N 3
FPGA_CDONE 4
PROTECTIVE_STATUS_FPGA 5
FPGA_ENDPOINT_READY 6
FPGA_MOTION_REQUEST 7
FPGA_PROGRESS_WDI 8
JOUT12001
INTERNAL BOUNDARY
TX_PERMIT 1
HW_WATCHDOG_OK_N 2
HW_PERMISSION_LATCHED 3
WD_HEALTH_RAW_N 4
LOGIC_GND 5

120pF:64.288ms nominal watchdog. NOT20ms.
100pF nominal would violate tested minimum at –tolerance.
Falling progress edge occurs every TWO accepted UPDATEs.
WDO auto–releases after170–230ms; heartbeat alone
cannot set the arm latch. SYNC + deliberate ARM required.
CDONE pullup belongs to FPGA block: no local pulldown.

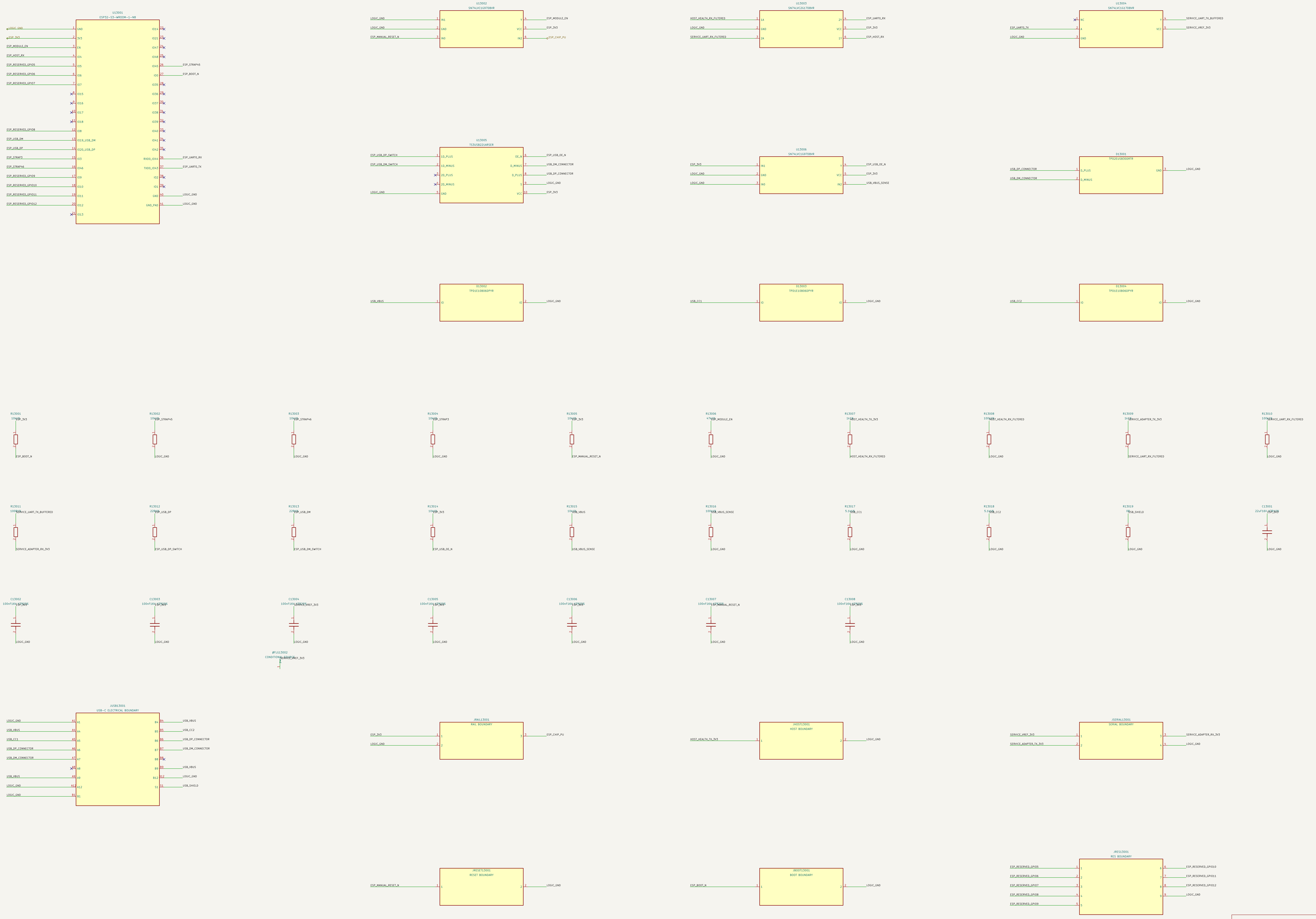
HW_WATCHDOG_OK_N –> RTL inhibit_n. Never return TX_PERMIT there.
Raw interlock has no endpoint_ready / motion_request / latch dependency.
D is permanently HIGH; clock is FPGA_MOTION_REQUEST rising edge.
Asynchronous clear is raw interlock AND endpoint_ready.
TX_PERMIT = latched_arm AND motion_request AND raw_interlock.

TX_PERMIT drives six AM26C31 G inputs + six10k pulldowns; /G fixedHIGH.
DefaultLOW disables transmitters. Per–drive active–low enables are separate.
EXT_PERMIT_3V3 is protected internal logic, never raw24V or certified E–stop.
Single–channel candidate: no safety certification, fault tolerance or controlled stop.
Brownout slew, partial power, metastability, wiring and physical timing remain open.
Clearing output permission can truncate a pulse; reconcile position before recovery.

Sheet: /machine_permission/
File: machine_permission.kicad_sch
Title: Independent watchdog / global permission
Size: A1 Date: 2026–09–28 **Rev: r0.1**
KiCad E.D.A. 10.0.4 Id: 13/14

ESP32–S3 SERVICE / RECOVERY CANDIDATE – NO MOTION OR ACCESSORY OUTPUT AUTHORITY

Self–powered USB2 device; USB VBUS never powers controller. USB connector is an electrical boundary pending exact pad mapping.
U2 is Schmitt AND: module EN = sequencer ESP_CHIP_PU AND manual–reset release. Never ground sequencer output.
Host RX is GPIO4 via powered–off buffer; default firmware leaves it disabled. UART0 adapter VREF powers only its outgoing buffer.


U13001
ESP32–S3–WROOM–1–N8
LOGIC_GND
ESP_3V3
ESP_MODULE_EN
ESP_HOST_RX
ESP_RESERVED_GPIO5
ESP_RESERVED_GPIO6
ESP_RESERVED_GPIO7
ESP_RESERVED_GPIO8
ESP_USB_DM
ESP_USB_DP
ESP_STRAP3
ESP_STRAP46
ESP_RESERVED_GPIO9
ESP_RESERVED_GPIO10
ESP_RESERVED_GPIO11
ESP_RESERVED_GPIO12
ESP_STRAP45
ESP_BOOT_N
ESP_UART0_RX
ESP_UART0_TX
GND
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19_USB_DM
IO20_USB_DP
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
IO0
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
RXD0_IO44
TXD0_IO43
IO2
IO1
GND_PAD
U13002
SN74LVC1G97DBVR
IN1
IN0
Y
VCC
IN2
ESP_MANUAL_RESET_N
ESP_CHIP_PU
U13003
SN74LVC2G17DBVR
HOST_HEALTH_RX_FILTERED
SERVICE_UART_RX_FILTERED
1A
2A
2Y
1Y
U13004
SN74LVC1G17DBVR
NC
A
SERVICE_UART_TX_BUFFERED
SERVICE_VREF_3V3
U13005
TS3USB221ARSER
ESP_USB_DP_SWITCH
ESP_USB_DM_SWITCH
1D_PLUS
1D_MINUS
2D_PLUS
2D_MINUS
OE_N
D_MINUS
D_PLUS
S
ESP_USB_OE_N
USB_DM_CONNECTOR
USB_DP_CONNECTOR
U13006
SN74LVC1G97DBVR
USB_VBUS_SENSE
D13001
TPD2EUSB30DRTR
D13002
TPD1E10B06DPYR
USB_VBUS
D13003
TPD1E10B06DPYR
USB_CC1
D13004
TPD1E10B06DPYR
USB_CC2
IO
R13001 10k1%
R13002 10k1%
R13003 10k1%
R13004 10k1%
R13005 10k1%
R13006 47k1%
R13007 1k1%
R13008 100k1%
R13009 1k1%
R13010 100k1%
R13011 100R1%
R13012 22R1%
R13013 22R1%
R13014 10k1%
R13015 10k1%
R13016 100k1%
R13017 5.1k1%
R13018 5.1k1%
R13019 0R
C13001 22uF16V X7R10%
C13002 100nF16V X7R20%
C13003 100nF16V X7R20%
C13004 100nF16V X7R20%
C13005 100nF16V X7R20%
C13006 100nF16V X7R20%
C13007 100nF16V X7R20%
C13008 100nF16V X7R20%
HOST_HEALTH_TX_3V3
SERVICE_ADAPTER_TX_3V3
SERVICE_ADAPTER_RX_3V3
USB_SHIELD
#FLG13002
CONDITIONAL_SOURCE
JUSB13001
USB–C ELECTRICAL BOUNDARY
A1
A4
A5
A6
A7
A8
A9
A12
B1
B4
B5
B6
B7
B8
B9
B12
S1
JRAIL13001
RAIL BOUNDARY
JHOST13001
HOST BOUNDARY
JSERIAL13001
SERIAL BOUNDARY
JRESET13001
RESET BOUNDARY
JBOOT13001
BOOT BOUNDARY
JRES13001
RES BOUNDARY

All footprints unassigned: no board layout or manufacture claim. USB switch D pins face connector; 1D pins face ESP; port2NC andSground.
Two independent5.1k CC resistors. SBU pins NC. NoPD/high–voltage mode. Local USB shield bond toLOGIC_GND is provisional; assess enclosure/EMC.
JRESET/JBOOT: normally–open dry contacts only. HoldBOOT low, holdRESET>=20ms, releaseRESET, keepBOOTlow>=3ms thenrelease.
JRES is reserved internal3.3V logic only; current firmware provides no accessory support. No5V/24V/unpowered external drive.
Module antenna outsidecarrieredge preferred; enclosureclearance>=15mm all directions. Shared2.4GHzWi–Fi/BLE, no5GHzorClassicBT.

Sheet: /esp_service/
File: esp_service.kicad_sch

Title: ESP service and recovery

Size: A1 | Date: 2026–09–28 | **Rev: r0.1**
KiCad E.D.A. 10.0.4 | Id: 14/14